

Double-Edged Nanobiotic Platform with Protean Functionality: Leveraging the Synergistic Antibacterial Activity of a Food-Grade Peptide to Mitigate Multidrug-Resistant Bacterial Pathogens

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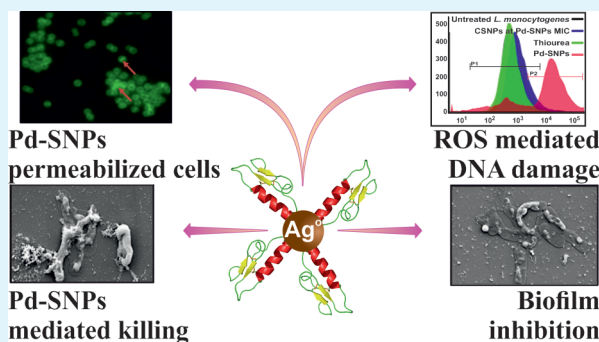
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ABSTRACT: While persistent efforts are being made to develop a novel arsenal against bacterial pathogens, the development of such materials remains a formidable challenge. One such strategy is to develop a multimodal antibacterial agent which will synergistically combat bacterial pathogens, including multidrug-resistant bacteria. Herein, we used pediocin, a class IIa bacteriocin, to decorate Ag⁰ and developed a double-edged nanoplatform (Pd-SNPs) that inherits intrinsic properties of both antibacterial moieties, which engenders strikingly high antibacterial potency against a broad spectrum of bacterial pathogens including the ESKAPE category without displaying adverse cytotoxicity. The enhanced antimicrobial activity of Pd-SNPs is due to their higher affinity with the bacterial cell wall, which allows Pd-SNPs to penetrate the outer membrane, inducing membrane depolarization and the disruption of membrane integrity. Bioreporter assays revealed the upregulation of *cpxP*, *degP*, and *sosX* genes, triggering the burst of reactive oxygen species which eventually cause bacterial cell death. Pd-SNPs prevented biofilm formation, eradicated established biofilms, and inhibited persister cells. Pd-SNPs display unprecedented advantages because they are heat-resistant, retain antibacterial activity in human serum, and alleviate vancomycin intermediate *Staphylococcus aureus* (VISA) infection in the mouse model. In addition, Pd-SNPs wrapped in biodegradable nanofibers mitigated *Listeria monocytogenes* in cheese samples. Collectively, Pd-SNPs exhibited excellent biocompatibility and *in vivo* therapeutic potency without allowing foreseeable resistance acquisition by pathogens. These findings underscore new avenues for using a potent biocompatible nanobiotic platform to combat a wide range of bacterial pathogens.

KEYWORDS: pediocin peptide, silver nanoparticles, synergistic antibacterial activity, antibiofilm activity, antipersister activity, nanofibers



INTRODUCTION

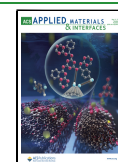
Antibiotics have paved the way for unprecedented medical and societal developments and thus are the most frequently prescribed drugs in modern times.¹ However, indiscriminate use and a dearth of novel therapeutics entering the antibiotic pipeline have led to an escalation of the crisis of antibiotic-resistant pathogens, ushering us into the imminent postantibiotic era.^{2,3} The present pandemic associated with the use of massive amounts of antimicrobials and sanitizers is likely to escalate the next pandemic of antibiotic-resistant pathogens.^{4,5} The resistance of pathogenic bacteria to drugs is mainly attributed to evolving multiple strategies to either efflux or inactivate the antibiotics, the formation of biofilms, and turning into persisters.^{6,7} A recent report estimated that, in addition to an economic loss of US \$100 trillion, antibiotic-resistant infections are expected to kill 10 million people each year by 2050.⁸ Myriad efforts have been dedicated to the battle against

resistant infections, leading to the development of new antibacterial materials which differ in mode of action from antibiotics.^{9–14} Antimicrobial peptides (AMPs) represent one such category which has been advocated as potential therapeutic alternative to antibiotics.¹⁵ The fact that these molecular knives act by inflicting severe damage to bacterial cell walls makes it difficult for the pathogen to evolve and acquire resistance against AMPs. The primary mechanism of action of AMPs is ascribed to electrostatic interaction with the target cell membrane, resulting in physical damage and the

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leakage of cellular contents.¹⁶ However, their widespread use is hindered by a narrow antibacterial spectrum, susceptibility to proteases, temperature sensitivity, high hemolysis, and cytotoxicity.^{17,18}

Besides AMPs, different metals in the nano form have also been studied for their ability to mitigate bacterial infections.^{19–21} Silver is a promising metal with a broad antibacterial spectrum and has been used in nanomaterials extensively.^{22,23} However, the risk of potential toxicity toward mammalian cells under antibacterial concentrations has impeded their widespread use in the food, cosmetics, and healthcare industries.²⁴ Thus, it is imperative to develop robust tools which do not show mammalian cell toxicity and are not amenable to developing resistance against bacterial pathogens.

The combinatorial therapy of antibacterial agents can synergistically inhibit the pathogens on multiple target fronts and thus may overcome the aforementioned problems.^{25–30} Pediocin-34, a small peptide bacteriocin (3.5 kDa) produced by a generally regarded as safe (GRAS) lactic acid bacterium *Pediococcus pentosaceus* is highly effective in controlling Gram-positive foodborne pathogen *Listeria monocytogenes*.³¹ To assess the synergy between silver and antimicrobial peptide pediocin-34 and to explore the feasibility to be delivered as a single package which attacks the bacteria from distinct fronts, we synthesized a double-edged sword: pediocin-34-functionalized silver nanoparticles (Pd-SNPs) as an antibacterial nanobiotic system which possesses the intrinsic properties of both bactericides. The developed Pd-SNPs displayed the functionality of a double-edged sword with synergistic bactericidal activity at extremely low dosages against both Gram-positive and Gram-negative bacterial pathogens. We deciphered the multimodal antimicrobial mechanism of action of Pd-SNPs and its effect on the mitigation of biofilm formation, persister development and resistance evolution. Pd-SNPs exhibited biocompatibility with mouse fibroblasts (NIH3T3), human embryonic kidney cells (HEK293), and human red blood cells (hRBCs) without displaying *in vivo* toxicity. Furthermore, the *in vivo* recovery of mice infection with vancomycin intermediate *Staphylococcus aureus* (VISA) verified the superlative therapeutic efficacy of Pd-SNPs. Additionally, we also demonstrate that Pd-SNPs can be incorporated with polylactic acid (PLA) to make nanofibers that pave the way to mitigating prominent foodborne pathogens.

EXPERIMENTAL SECTION

Materials. Silver nitrate (AgNO_3), sodium borohydride (NaBH_4), De Man, Rogosa, and Sharpe (MRS) broth, Luria–Bertani (LB) broth, Muller Hinton (MH) broth, and brain heart infusion (BHI) media were purchased from Merck, Germany. A LIVE/DEAD BacLight bacterial viability kit, BODIPY FL vancomycin, 2,7-dichlorofluorescein diacetate (DCFH-DA), and bis-1,3-dibutylbarbituric acid trimethine oxonol [DiBAC4(3)] were procured from Thermo Fisher Scientific (USA). *N*-Dimethylformamide (DMF) and 3,3'-dipropylthiadicarbocyanine iodide [DiSC3(5)] were purchased from Sigma-Aldrich (USA). All cell culture media and fetal bovine serum (FBS) were purchased from Gibco-BRL (USA). A terminal deoxynucleotidyl transferase-mediated dUTP nick end labeling (TUNEL) assay kit was obtained from Roche, Molecular Biochemicals (Germany). Bacterial strains *Listeria monocytogenes*, *Escherichia coli* O157:H7 (ATCC 35150), *Salmonella typhi* (ATCC 14901), *Staphylococcus aureus* (ATCC 29213), and *Acinetobacter baumannii* AYE (ATCC BAA-1710, resistant to 16 drugs) were obtained from the American Type Culture Collection (ATCC) (USA). Pediocin-producing *Pediococcus pentosaceus* and its indicator

strain *Pediococcus acidilactici* were isolated in the laboratory of Dr. Ravinder Kumar Malik of the National Dairy Research Institute (Karnal, India). The clinical isolate of *Staphylococcus aureus* (vancomycin intermediate *Staphylococcus aureus*; VISA-ST1745) was procured from Sanjay Gandhi Post Graduate Institute of Medical Science (SGPGIMS) (Lucknow, India). Swiss albino mice were procured from NIPER (Mohali, India) and were maintained in a small animal facility at 22 ± 3 °C with a relative humidity of 50–70% and light and dark cycles of 12:12 h. Ultrapure water (18.2 Ω M cm, Millipore) was used to prepare nanoparticle solutions.

Production and Purification of Pediocin. *Pediococcus pentosaceus* cells were incubated in MRS broth at 37 °C under static conditions overnight. The overnight-grown cells were subcultured (1% inoculum) in 500 mL of fresh MRS broth for 36 h at 37 °C under static conditions. Then the bacterial cells were centrifuged at 10 000 rpm for 15 min at 4 °C, and the supernatant was subjected to precipitation with 80% ammonium sulfate at 4 °C for 6 h. The precipitate was pelleted down by centrifugation at 10 000 rpm for 30 min, and the cells were finally resuspended in a minimum volume of 1× phosphate-buffered saline (PBS). The precipitated ammonium sulfate sample was dialyzed against 1× PBS buffer and lyophilized until only a dry residue remained. For subsequent purification using reverse-phase high-performance liquid chromatography (RP-HPLC, Waters, USA), the lyophilized sample was resuspended in 1 mL of 1× PBS buffer, mixed with sample buffer (20% acetonitrile + 2% trifluoroacetic acid) in a ratio of 1:3 (sample buffer/pediocin), and centrifuged for 5 min at 2000 rpm. The supernatant was loaded onto a C18 column (Sunfire C18 5 μ m, 4.6 mm $\times$ 250 mm), which was pre-equilibrated with mobile phase A (0.1% trifluoroacetic acid in water). The conditions for the gradient elution of solvent B (0.1% trifluoroacetic acid in acetonitrile) against solvent A (0.1% trifluoroacetic acid in water) at a 1 mL/min flow rate were as follows: 40–120 min, 40% solvent A and 60% solvent B; 120–125 min, 100% solvent B; and 125–130 min, 100% solvent A. The size and homogeneity of the peptide were determined using SDS-PAGE chromatography followed by silver staining. The collected fractions (76–106 min) were pooled and freeze-dried. The spot assay was used to measure the activity unit (AU) against *P. acidilactici* LB42, an indicator strain of pediocin.

Synthesis and Characterization of Pediocin-Capped Silver Nanoparticles (Pd-SNPs). Before the preparation of NPs, the glassware that was used was cleaned using freshly prepared aqua regia solution (3:1 of concentrated HCl and HNO_3), rinsed thoroughly in Milli-Q water, and oven-dried for 12 h. Pd-SNPs were synthesized by silver nitrate (AgNO_3) reduction in the presence of purified pediocin and sodium borohydride (NaBH_4) as follows: 8.8 mL of Milli-Q water was taken into a 50 mL glass beaker standing on a magnetic stirrer. AgNO_3 (20 mM, 1 mL) was added slowly to the beaker with purified pediocin (1600 AU/mL) under stirring. Thereafter, 100 μ L of 60 mM NaBH_4 solution was consequently added dropwise to the solution until the solution became a vivid dark yellow, and it was stirred vigorously for 30 min in the presence of white light (40 W, 141.3 lux), resulting in the formation of Pd-SNPs. Furthermore, to remove the unreacted precursors (pediocin, AgNO_3 , and NaBH_4), the as-synthesized Pd-SNPs were centrifuged twice at 20 000 rpm for 20 min and washed twice with Milli-Q water. The Pd-SNPs suspension was kept at 4 °C for further use. The fluorescent OPA test was used to measure the quantity of pediocin on the functionalized NPs and their stability in terms of pediocin leaching.³² Furthermore, the stability of the produced Pd-SNPs kept at 4 °C was also evaluated by monitoring the zeta potential and antibacterial activity weekly for 2 months. Citrate-capped silver nanoparticles (CSNPs) were also synthesized under the same conditions except that purified pediocin was replaced by 100 μ L of 15 mM trisodium citrate and were used for the control experiments.

To confirm the synthesis of Pd-SNPs, UV–vis absorption spectra were acquired in the range of 200–700 nm using a Synergy-H1 microplate reader from BioTek (USA). The morphology and size of the particles were analyzed by transmission electron microscopy (TEM; JEOL, Japan). TEM samples were prepared by drop-casting

and air-drying the nanoparticles on a carbon-coated copper grid (200 mesh) and were observed at 200 keV. The average hydrodynamic diameter (R_h) and zeta potential of the NPs were analyzed using dynamic light scattering (DLS, Zetasizer Nano-ZS, Malvern Instruments Ltd., U.K.). Fourier transform infrared spectra (FTIR, Thermo Nicolet NEXUS 670, USA) of CSNPs and Pd-SNPs were acquired in the 400–4000 cm^{-1} frequency range. Nuclear magnetic resonance spectra (^1H NMR, AVANCE III AV 500, Bruker, USA) for both pediocin and Pd-SNPs at a concentration of 200 $\mu\text{g}/\text{mL}$ (based on the peptide concentration) were recorded on a TXI probe. The experiments were performed in 5 mm NMR tubes with 550 μL of the samples dissolved in water (90% H_2O + 10% D_2O). The following parameters were used: spectral sweep width, 16 ppm; data points, 16K; total relaxation delay (RD), 5 s; and number of scans, 256. Each FID was zero-filled to 4096 data points before FT, and a sine-bell apodization function was used. The spectra were manually baseline-corrected, phase-corrected, and referenced. The determination of silver ions (Ag^+) in CSNPs and Pd-SNPs was carried out with an atomic absorption spectrophotometer (AAS, GBC Scientific Equipment, USA) using silver nitrate standard solutions.

Determination of the Minimum Inhibitory Concentration (MIC) and Fractional Inhibitory Concentration (FIC). The MIC of as-synthesized NPs (CSNPs and Pd-SNPs) was evaluated by the broth microdilution method of the Clinical and Laboratory Standards Institute (CLSI, USA) in BHI and LB broth for Gram-positive and Gram-negative bacterial strains, respectively.³³ MIC was implied to be the lowest concentration of Pd-SNPs inhibiting the observable bacterial growth after 10 h of incubation. As a control, CSNPs with the same silver concentration were used. Each experiment was performed in duplicate and repeated three times. The overnight-grown microorganisms were subcultured in their respective broths and grown until $\text{OD}_{600\text{ nm}} = 0.4$ was reached. Furthermore, each bacterial suspension was diluted 1000-fold to attain an initial inoculum of 10^5 colony-forming units/mL (CFU/mL) and mixed with an equal volume of 2-fold-diluted nanoparticle solution, and the cellular growth was observed at $\text{OD}_{600\text{ nm}}$ after 10 h of incubation at 37 °C under static conditions in a humidity-controlled incubator (Kuhner LT-X, Switzerland). Growth above 10% at $\text{OD}_{600\text{ nm}}$ of the positive control (bacterial suspension without nanoparticles) was considered to be uninhibited growth compared to the uninoculated broth solution which was used as a negative control. We determined the MICs of Pd-SNPs and CSNPs against *Listeria monocytogenes*, vancomycin intermediate *Staphylococcus aureus* (VISA-ST1745), *Bacillus cereus* (ATCC 11778), *Enterococcus faecium*, *Vibrio fluvialis* L-15318, *Klebsiella pneumoniae* (ATCC 700603), *Pseudomonas aeruginosa* (ATCC 27853), *Acinetobacter baumannii* AYE (ATCC BAA-1710), *Clostridium perfringens* (ATCC 3624), *Salmonella enterica* subsp. *enterica* serovar *typhi* (ATCC 14901), and *Escherichia coli* O157:H7 (ATCC 35150). The *in vitro* synergistic antibacterial activities of CSNPs (compound A) and pediocin (compound B) were determined using the FIC index (FICI). The FIC index was calculated as follows:

$$\text{FIC index} = \frac{\text{MIC of compound A in combination}}{\text{MIC of compound A alone}} + \frac{\text{MIC of compound B in combination}}{\text{MIC of compound B alone}} \quad (1)$$

The interaction of pediocin and NPs was defined as synergistic action if the FIC index was ≤ 0.5 .³⁴

Interaction Studies of Pd-SNPs with Bacterial Membrane Proteins. To investigate the interaction of Pd-SNPs with bacterial membranes, outer membrane (OM) fractions of *V. fluvialis* L-15318 and *L. monocytogenes* were isolated using the N-lauroyl sarcosinate method with slight modifications.³⁵ *L. monocytogenes* and *V. fluvialis* L-15318 cells were harvested (*L. monocytogenes* cell pellet was incubated with 2 mg/mL lysozyme at 25 °C for 20 min) and rinsed once with 1× PBS and diluted to 10^5 CFU/mL in 1× PBS supplemented with 0.1 mM phenylmethylsulfonyl fluoride (PMSF). The resuspended cells were disrupted by ultrasonication with the

amplitude set at 30% with a pulse sequence of 5 s on and 10 s off for 15 min (Branson Digital Sonifier SFX 550, USA). The resulting extract was pelleted by centrifugation at 10 000 rpm for 10 min at 4 °C, and the supernatant was incubated with 2% sodium N-lauroyl sarcosinate (w/v) in 1× PBS for 30 min at 25 °C. The OM fractions were collected by ultracentrifugation at 90 000 rpm for 1 h at 4 °C (Beckman Coulter, USA), washed once with 2% sodium N-lauroyl sarcosinate (wt/vol), collected as described above, resuspended in 1× PBS, and stored at −20 °C. OM fractions of *L. monocytogenes* and *V. fluvialis* L-15318 (50 μg each) were incubated with different concentrations of Pd-SNPs and CSNPs for 30 min in 10 mM HEPES buffer, pH 7.2. Fluorescence spectra of the reaction mixture were recorded using a Synergy H1 microplate reader from BioTek (USA) in the range of 300–500 nm with an excitation wavelength of 290 nm. Furthermore, the concentration-dependent quenching of the fluorescence of proteins was investigated by the following Stern–Volmer equation³⁶

$$\frac{F_0}{F} = 1 + K_{sv}[Q] \quad (2)$$

where $[Q]$ is the concentration of quencher (SNPs) and K_{sv} is a quenching constant. F_0 and F are the steady-state fluorescence intensities in the absence and presence of quencher, respectively.

Effect of Pd-SNPs on *L. monocytogenes* Morphology Determined by Scanning Electron Microscopy. To visualize the morphological effects of Pd-SNPs, the treated bacterial cells were imaged using a field-emission scanning electron microscope (FE-SEM, ultra plus field emission; Carl Zeiss, Germany). The bacterial cells (10^5 CFU/mL) were incubated with the MIC values of Pd-SNPs, CSNPs, and Pd-SNPs for 2 h at ambient temperature. Following incubation, cells were harvested by centrifugation at 5000 rpm at 4 °C and were resuspended in 1× PBS. A thin smear of treated and untreated cells was primarily fixed on glass slides with 2.5% glutaraldehyde for 30 min, followed by gradual dehydration with a series of aqueous ethanol gradient solutions (25, 50, 70, 80, 90, and 100%). The prepared samples were air-dried, followed by sputtering deposition of the gold layer, and were observed under SEM.

Localization of Fluorescently Labeled Vancomycin in *Vibrio fluvialis* and VISA Cells. To understand the bactericidal mechanism of action of Pd-SNPs, BODIPY FL vancomycin was used to localize in Pd-SNP-treated bacterial cells. For this, 10^5 CFU/mL of *V. fluvialis* L-15318 and VISA-ST1745 cells were treated with 1× MIC of Pd-SNPs along with 1 μM BODIPY FL vancomycin and were incubated in the dark for 3 h at 37 °C. After incubation, the cells were pelleted by centrifugation at 5000 rpm for 5 min and resuspended in 1× PBS. Only vancomycin-treated cells were used as a control.

Viability Analysis of Pd-SNPs-Treated *Listeria monocytogenes*. To examine the bactericidal effect of Pd-SNPs, *L. monocytogenes* cells (10^5 CFU/mL) were treated with the MIC (2.6 Ag^0 $\mu\text{g}/\text{mL}$) of bath-sonicated Pd-SNPs and with CSNPs at the MIC of Pd-SNPs, whereas untreated cells were taken as the negative control. After 2 h, cells were collected by centrifugation at 10 000 rpm for 10 min at 4 °C and were resuspended in 100 μL of 1× PBS. MIC-treated and untreated samples were stained for 20 min with a mixture of a fluorescent dye solution of a live/dead staining kit and were viewed under an inverted fluorescence microscope at 1000× magnification (Axio Scope A1; Carl Zeiss, Germany).

Time-Kill Kinetics. For time-kill experiments, the growth kinetics of bacterial cells (*L. monocytogenes*, *V. fluvialis* L-15318, and VISA-ST1745) were studied via a viable cell-counting method. Each bacterial cell (10^5 CFU/mL) was incubated with Pd-SNPs (at the MIC value) and CSNPs (at the MIC value of Pd-SNPs) under the same conditions as for the MIC assay described above. As an untreated control, only bacterial cells were incubated in the respective broth medium. At different time intervals ($t = 0, 1, 3, 6, 9$, and 12 h), treated and untreated samples were withdrawn, serially diluted using sterile 1× PBS, and plated on the respective media agar plates. After 12 h of incubation at 37 °C, viable colonies (CFU/mL) were counted and the kill kinetic curve was plotted.

Outer and Inner Membrane Permeability Assay. To test the ability of Pd-SNPs to permeabilize the inner (IM) and outer membranes (OM) of Gram-negative bacteria, *E. coli* ML-35p [(a lactose permease-deficient species with periplasmic β -lactamase and constitutive β -galactosidase activity (i^- , y^- , z^+)] in 96-well microtiter plates was used as stated in the previously described procedure.³⁷ The overnight-grown cells were subcultured in LB medium containing 100 μ g/mL ampicillin, rinsed twice, and resuspended in 10 mM HEPES buffer to 10^5 cells/mL. This bacterial suspension (50 μ L) was added with 0.5 $\times$ MIC and 1 $\times$ MIC of Pd-SNPs and CSNPs at the MIC value of Pd-SNPs. Furthermore, 50 μ L of nitrocefin (NCF; for OM permeabilization) solution (60 μ g/mL stock) or 50 μ L of *O*-nitrophenyl- β -D-galactopyranoside (ONPG; for IM permeabilization) solution (300 μ g/mL stock) was added and a total volume of 150 μ L was made up by 10 mM HEPES buffer, pH 7.2. Polymyxin B (5 μ M, for *E. coli* membrane permeabilization)³⁸ and wells without any SNPs were used as positive and negative controls, respectively. The permeabilization of OM and IM was monitored at $\lambda_{500\text{ nm}}$ and $\lambda_{420\text{ nm}}$, respectively, for 1 h at a 10 min interval on a microplate reader.

Membrane Depolarization Assay. The ability of Pd-SNPs to perturb the membrane potential was discerned using membrane-potential-sensitive fluorophores DiBAC4(3) (for *L. monocytogenes* and *V. fluvialis* L-15318) and DiSC3(5) (for VISA-ST1745) as reported earlier.^{39,40} Briefly, the diluted bacterial cells (10^5 CFU/mL) were challenged with 0.5 $\times$ MIC, 1 $\times$ MIC, and 2 $\times$ MIC of Pd-SNPs and CSNPs at the MIC of Pd-SNPs. After 2 h of incubation, DiBAC4(3) (10 μ g/mL) and DiSC3(5) (0.5 μ M) were incubated with respective cell suspensions for 30 min, following which the readings were recorded for DiBAC4(3) (excitation at 490 nm and emission at 516 nm) and DiSC3(5) (excitation 622 nm and emission at 670 nm). Protonophore carbonyl cyanide chlorophenyl hydrazine (CCCP) and 1 $\times$ PBS buffer were used only as positive and negative controls, respectively.

Reactive Oxygen Species (ROS) Generation Assay. The intracellular ROS generation of cells was investigated using the nonfluorescent molecule DCFH-DA, which is deacetylated by cellular esterases to DCFH that can interact with ROS to generate fluorescent 2',7'-dichlorofluorescein (DCF).⁴¹ Briefly, the bacterial cells of *L. monocytogenes*, *V. fluvialis* L-15318, and VISA-ST1745 (10^5 CFU/mL each) were incubated in the dark with 100 μ M DCFH-DA at 37 $^\circ$ C for 30 min. Upon incubation, cells were centrifuged at 5000 rpm at 4 $^\circ$ C, rinsed with 1 $\times$ PBS to remove extra DCFH-DA, and incubated with 0.5 $\times$ MIC, 1 $\times$ MIC, and 2 $\times$ MIC of Pd-SNPs for 2 h in the dark. The bacterial cells incubated with 1 $\times$ PBS and 10 μ M hydrogen peroxide (H_2O_2) were used as negative and positive controls, respectively. The fluorescence was measured immediately after light exposure by excitation at 485 nm and emission at 530 nm for 1 h at an interval of every 10 min on a microplate reader.

Determination of DNA Fragmentation Using the TUNEL Assay. DNA fragmentation in *L. monocytogenes* was quantified with a terminal deoxynucleotidyl transferase-mediated dUTP nick end-labeling (TUNEL) assay and was performed according to the manufacturer's instructions (Roche Molecular Biochemicals, Germany). For DNA fragmentation studies, *L. monocytogenes* cells (10^5 CFU/mL) were preincubated for 2 h with 0.5 $\times$ MIC of Pd-SNPs and were rinsed three times in 1 $\times$ PBS. The cells were fixed with 4% formaldehyde for 2 h at 37 $^\circ$ C. After washing with 1 $\times$ PBS, cells were treated with 0.1% Triton X-100 and 0.1% sodium citrate for 2 min at 4 $^\circ$ C and washed again with 1 $\times$ PBS. The permeabilized cells were incubated in the dark for 1 h at room temperature with 50 μ L of TUNEL reagent. Upon incubation, the cells were rinsed twice with 1 $\times$ PBS and were analyzed by flow cytometry (BD FACS Verse BD Biosciences, USA) with excitation at 488 nm and emission at 527–532 nm. The cells without nanoparticle treatment were used as a negative control. The effect of the ROS quencher on DNA damage was studied using thiourea (100 mM).

Mechanism of Action Studies Using a Whole Cell Biosensor Array. The responses of stress-responsive fluorescent bioreporters in *E. coli* BW25113 to Pd-SNPs and CSNPs were assayed (Table S3).

Overnight bacterial cells were diluted 100-fold in 6 mL of culture media (5 g/L NaCl, 2 g/L glucose, 10 g/L tryptone, 11.9 g/L HEPES) and were grown to an OD of 0.4. Tetracycline was used as a selection marker for proper plasmid maintenance. Culture aliquots (190 μ L) were then transferred to a black 96-well clear-bottom microtiter plate, and the volume was adjusted to 200 μ L with Pd-SNPs and CSNPs at a sub-MIC concentration of silver ions. Finally, the fluorescence (485 nm excitation/525 nm emission) and absorbance (600 nm) of the samples were measured every hour, and incubation was carried out at 37 $^\circ$ C for 8 h.

Biofilm Inhibition Assay. The inhibition of biofilm formation by Pd-SNPs was determined and quantified by crystal violet staining.⁴² Briefly, the overnight grown *L. monocytogenes* and *V. fluvialis* L-15318 cells were subcultured in BHI and LB medium and grown until the OD_{600 nm} reached 1.0. The above cultures were diluted 1:10 in the respective media, and 100 μ L of this diluted culture was incubated (in triplicate) with 100 μ L of the respective medium with varying concentrations of CSNPs (0–133.2 μ g/mL) and Pd-SNPs (0–20.8 μ g/mL) at 37 $^\circ$ C for 24 and 48 h in a humidity-controlled incubator under static conditions. The subcultured diluted cells without nanoparticles and the respective broth alone were used as positive and negative controls, respectively. Following incubation, the wells were washed three times with sterile 0.9% saline to remove planktonic bacterial cells. The biofilm was stained with 300 μ L of crystal violet solution (0.1% w/v) for 15 min, followed by thorough washing, drying, and finally dissolution in 300 μ L of 30% acetic acid solution. The inhibition of the biofilm was assessed by quantifying the OD_{550 nm} using a Synergy H1 microplate reader.⁴³

Visualization of the Biofilm Inhibition and Eradication Ability of Pd-SNPs Using Scanning Electron Microscopy. To investigate the antibiofilm activity of Pd-SNPs, *L. monocytogenes* (as described above) was seeded in a six-well plate with a sterile coverslip along with the MBIC value of Pd-SNPs and incubated for 48 h at 37 $^\circ$ C in a humidity-controlled incubator under static conditions. Following the incubation, the samples were rinsed twice with 1 $\times$ PBS to eliminate the nonadherent planktonic cells and were prepared for SEM analysis. To study the biofilm removal activity, the aliquots of 1:10 diluted culture were placed in a six-well plate and incubated as described above. After biofilm formation, the nonadherent cells were washed twice with 1 $\times$ PBS. The samples were then treated with 5 $\times$ MBIC of Pd-SNPs for 3 h at 37 $^\circ$ C under static conditions. The samples were prepared for SEM analysis to visualize the biofilm eradication ability of Pd-SNPs.

Bactericidal Activity against Persister Cells. To develop persister cells from a biofilm culture, 100 μ L of *V. fluvialis* L-15318 and VISA-ST1745 cells (10^8 CFU/mL) was incubated in a 96-well polystyrene plate at 37 $^\circ$ C in a humidity-controlled incubator under static conditions for 24 h. The wells were rinsed twice with 1 $\times$ PBS to remove nonadherent planktonic cells, and biofilms were exposed to fresh broth containing antibiotics separately (vancomycin and colistin at 100 $\times$ MIC). After incubation for 24 h, nonadherent cells were washed away with 1 $\times$ PBS, and adherent bacterial cells were released in 1 $\times$ PBS by 5 min of sonication and were incubated with Pd-SNPs at different MICs. The bacterial cells treated with 1 $\times$ PBS without Pd-SNPs were considered to be a control. Bacterial viability was determined by CFU counting, after 2 and 4 h of incubation at 37 $^\circ$ C.

Assessment of the Heat Stability of Pd-SNPs. To examine the effect of heat treatment on the antibacterial activity of the Pd-SNPs, a nanobioconjugate was heat-treated at 95 $^\circ$ C for 10 and 30 min. After the incubation, Pd-SNPs at 2 $\times$ MIC were added to 100 μ L of BHI broth in a microtiter plate with 100 μ L of diluted *L. monocytogenes* cells at a concentration of 10^5 CFU/mL in BHI broth. The untreated Pd-SNPs at a concentration of MIC were taken as a positive control. The microtiter plate was incubated at 37 $^\circ$ C in a humidity-controlled incubator under static conditions, and the growth of bacteria was estimated by enumerating CFU in the agar plate. A bar graph was plotted to evaluate the antibacterial activity of heat-treated Pd-SNPs as compared to the untreated Pd-SNPs. To visualize the effect of temperature on the Pd-SNP morphology, TEM images were taken and compared to those of untreated Pd-SNPs.

Stability Assessment of Pd-SNPs in Human Serum and Plasma. To determine the stability of nanoparticles in human serum and plasma, we checked the antibacterial activity of Pd-SNPs in human plasma and serum. *Listeria monocytogenes* (10^5 cells, 100 μ L) was incubated with Pd-SNPs at 1 $\times$ MIC and 2 $\times$ MIC concentrations supplemented with human serum and plasma to a final volume of 200 μ L. Following the incubation for 10 h under static conditions in a humidity-controlled incubator at 37 $^{\circ}$ C, the growth of bacteria was assessed by measuring the CFU/mL in the agar plate.

Assessment of Resistance Development. The development of resistance to the Pd-SNPs was assessed with some modifications to the earlier described method.⁴⁴ In a 96-well plate, 100 μ L of *L. monocytogenes* and VISA-ST1745 (10^5 cells each) was added to 100 μ L of Pd-SNPs/pedocin solution in BHI broth and was incubated for 10 h at 37 $^{\circ}$ C in a humidity-controlled incubator under static conditions. The MIC, the lowest concentration of Pd-SNPs/pedocin that prevents the visible growth of bacteria, was determined. Thereafter, 5 μ L of the bacterial growth from 0.5 $\times$ MIC suspension was diluted again and transferred to a fresh medium supplemented with Pd-SNPs and incubated as described above. This process was repeated for 20 serial passages for *L. monocytogenes* and VISA-ST1745. As a control, MIC values were also obtained using fresh *L. monocytogenes* and VISA-ST1745 cells serially passaged in BHI broth alone.

In Vitro Cytocompatibility Assay. The *in vitro* cytocompatibility of Pd-SNPs with HEK293 (human embryonic kidney cells) and NIH 3T3 (mouse fibroblast cells) cells was evaluated with the MTT assay.⁴⁵ Both the HEK293 and NIH 3T3 cells were cultured in Dulbecco's Modified Eagle's Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) at 37 $^{\circ}$ C with 5% CO₂ in an incubator. The cells were seeded in each well of a 96-well microtiter plate at a density of 5000 cells/well in triplicate and incubated for 24 h. The cells were then exposed to different concentrations of Pd-SNPs (0–12.5 μ g/mL) and CSNPs (0–12.5 μ g/mL) for 4 h. To each well, 10 μ L of MTT (5 mg/mL) solution was added and incubated for another 4 h to form formazan. Following the incubation, the medium was aspirated out, and the purple insoluble formazan crystals that formed in the well were solubilized in 200 μ L of dimethyl sulfoxide (DMSO). The absorbance in each well was measured at 570 nm and was used to estimate the cell viability relative to the control cells. The absorbance of the cells without nanoparticles in the culture was considered to be 100% viability as a reference. The cell viability was expressed as the absorbance relative to that of the control.

Hemolytic Activity Assay. To assess the extent of toxicity of Pd-SNPs, a hemolytic evaluation was performed.⁴⁶ Fresh human whole blood was drawn as approved by the Institutional Human Ethics Committee (BT/IHEC-7/2020/7866) from a healthy donor in a sterile vacutainer in the affiliated hospital of the Indian Institute of Technology (Roorkee, India). The red blood cells (RBC) were separated from the blood sample by centrifugation at 1500g for 5 min and washed three times with sterile 0.9% saline solution. The final pellet was diluted 1:9 (v/v) in sterile 0.9% saline solution and subsequently diluted 1:24 (v/v) in sterile 1 $\times$ PBS. RBC suspensions were mixed with different concentrations of Pd-SNPs (0–25 μ g/mL) and CSNPs (0–33 μ g/mL) and incubated at 37 $^{\circ}$ C for 4 h, followed by centrifugation at 1500g for 5 min. The supernatant (200 μ L) was transferred to a 96-well plate, and hemoglobin release was recorded at 576 nm on a Synergy-H1 microplate reader. The RBC suspension was incubated in 1 $\times$ PBS and Triton-X100 (0.1% v/v) and was used as a negative and positive control, respectively. For each concentration and control, the experiment was performed in triplicate. The percent hemolysis was calculated using the following equation:

$$\text{hemolysis \%} = \frac{\text{OD}_{576 \text{ nm}}(\text{treated sample}) - \text{OD}_{576 \text{ nm}}(\text{negative sample})}{\text{OD}_{576 \text{ nm}}(\text{positive sample}) - \text{OD}_{576 \text{ nm}}(\text{negative sample})} \times 100\%$$

Bacterial Invasion Assay. The bacterial invasion assay was carried out using the HEK293 cell line. The cell line was cultured using DMEM medium supplemented with 10% (v/v) fetal calf serum.

The *S. aureus* strains SA-1199B, VISA-ST1745, and SA-K1758 were used for the invasion assay. The cell monolayers of HEK293 were incubated at 37 $^{\circ}$ C with 5% CO₂ for 48 h to reach a confluence of nearly 2×10^5 cells per well. HEK293 cells (2×10^5 cells) were infected with bacteria by adding 10^6 CFU/well (multiplicity of infection = 10) of freshly grown bacteria along with Pd-SNPs and were incubated for 1 h at 37 $^{\circ}$ C with 5% CO₂. After 1 h, the infected HEK 293 cells were rinsed three times with 1 $\times$ PBS and DMEM medium supplemented with 5 μ g of lysostaphin/mL (AMBI, Inc., USA) was added to each well to lyse extracellularly adherent *S. aureus*. CFU enumeration on agar plates was then used to establish the absence of extracellular bacteria. Intracellular bacteria were enumerated by host-cell lysis using 0.25% (w/v) SDS and plated onto trypticase soy agar.

In Vivo Efficacy of Pd-SNPs in the VISA-ST1745 Infection Mouse Model. All animal experiments were conducted in compliance with the Institutional Animal Ethics Committee of the Indian Institute of Technology (Roorkee, India) (BT/IAEC/2017-20/06) according to the guidelines of the Committee for the Purpose of Control and Supervision of Experiments on Animals (Government of India, New Delhi, India). Forty-two female Swiss albino mice were acclimatized for 7 days and randomly grouped 2 days before the start of the experiment. On day 0 (at the start of the experiment), mice were infected with a bacterial strain via an intravenous route and 24 h postinfection, and treatment with Pd-SNPs was continued for 7 days through oral administration. The mice were divided ($n = 6$) into a negative control (vehicle), a positive control (vancomycin 10 mg/kg intravenous), and treatment groups (doses of 1, 2.5, 5, and 10 mg/kg of body weight for Pd-SNPs) as well as CSNPs at a higher dose of 10 mg/kg. The infection was transmitted through the intravenous route with 1×10^6 CFU/mL of VISA-ST1745. The treatment was given for 7 days postinfection, following which the animals were euthanized and the spleen, lung, liver, and kidneys were aseptically removed and homogenized. The bacterial load in these organs was enumerated using the plate dilution method.

In Vivo Toxicity Assessment of Pd-SNPs. The *in vivo* toxicity testing was performed in Swiss albino mice in accordance with the Organization for Economic Co-operation and Development's (OECD) test guideline 423. Briefly, three groups of six female mice each were taken; group I served as the vehicle control (healthy and uninfected), groups II and III were administered Pd-SNPs at two different concentrations (25 and 50 mg/kg of body weight) for 10 days. Furthermore, the same experiment was also executed in the animals infected with VISA-ST1745, along with the administration of Pd-SNPs at 50 mg/kg of body weight for 10 days. The animals were monitored for variations in body weight, and blood was collected for the determination of various physiological parameters. The animals were euthanized to examine gross necropsy, and the relative organ weights were also measured. The serum was examined for liver function (ALT, AST, ALKP, and tBIL) and lipid metabolism (total cholesterol) using an autoanalyzer (Kearneysville, USA).

Synthesis and Characterization of Biodegradable Pd-SNPs–Polylactic Acid (Pd-SNPs-PLA) Nanofibers. Pd-SNPs-PLA nanofibers were fabricated by electrospinning (ESPIN-NANO-PECO, India). Polylactic acid (PLA, 8%, Natur-Tec India Pvt. Ltd.) was dissolved in a 3:1 ratio of dichloromethane (DCM) and dimethylformamide (DMF). Furthermore, the as-prepared Pd-SNPs were centrifuged at 20 000 rpm for 20 min, and the pellet was resuspended in the above solution and kept under constant stirring overnight to obtain a homogeneous solution. The resultant mixture was placed in a 2.5 mL syringe for electrospinning to synthesize nanofibers on aluminum foil through a metal needle that was maintained 15 cm away from the collector (aluminum foil). The solution was supplied at a rate of 1.2 mL/h between the needle tip and the grounded collector at a fixed voltage of 13 kV.

The microstructure and the size of the nanofibers were determined by FE-SEM and TEM. The interaction of Pd-SNPs and PLA was investigated using FTIR spectroscopy. The samples were recorded in the 400–4000 cm^{−1} wavenumber range in the attenuated total reflectance (ATR) mode. Surface wettability in terms of the water

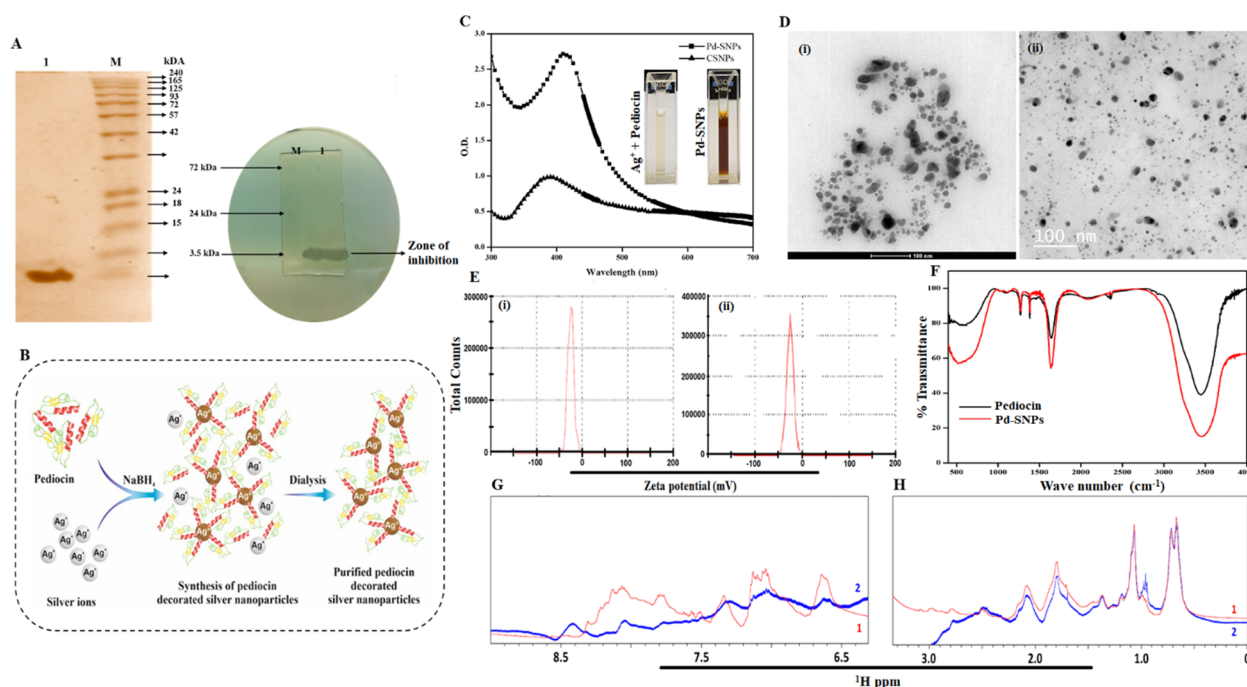


Figure 1. Synthesis and characterization of Pd-SNPs. (A) Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (15%, SDS-PAGE), followed by silver staining of purified pediocin peptide from *Pediococcus pentosaceus* with (lane 1) purified pediocin, (lane M) a broad-range protein ladder, and an in-gel overlay antibacterial assay showing a zone of inhibition against an indicator strain. (B) Schematic illustration of the synthesis of pediocin-capped silver nanoparticles (Pd-SNPs). (C) UV–vis absorption spectrum of an aqueous solution of CSNPs (▲) and Pd-SNPs (■). (D) TEM images of (i) CSNPs and (ii) Pd-SNPs. (E) Dynamic light scattering spectra of (i) CSNPs and (ii) Pd-SNPs. (F) Comparison of Fourier transform infrared spectra of pediocin and Pd-SNPs. ^1H nuclear magnetic resonance spectra of (G) the NH region and (H) the aliphatic proton region, with pediocin only (1, red) and peptide bound to the nanoparticles (2, blue).

contact angle was measured using a contact angle analyzer (Kruss DSA 100 easy drop, Hamburg, Germany). A water droplet was placed on the surface of a nanofiber mat and photographed immediately, and the contact angle value was measured. Three measurements were made on each sample, and the average values were calculated.

Disc Diffusion Antibacterial Assay of Pd-SNPs-PLA Nanofibers. The antimicrobial activity of a Pd-SNPs-PLA nanofibrous mat against *L. monocytogenes* was investigated qualitatively with a diffusion assay. An overnight culture of *L. monocytogenes* was subcultured in BHI broth until $\text{OD}_{600\text{ nm}}$ reached 0.2. The culture (100 μL) was spread on a BHI agar plate supplemented with 0.01% Triton-X100. A 1 cm $\times$ 1 cm Pd-SNPs-PLA nanofiber film was placed on the BHI agar plate and incubated at 37 $^\circ\text{C}$ for 24 h. Upon incubation, the inhibition zone diameter of the inoculated plate was measured. Only a PLA nanofiber mat was used as a control.

Silver Ion Release Kinetics. Silver-ion release from the SNP-PLA nanofiber film was determined using inductively coupled plasma mass spectrometry (ICPMS, ELAN DRC-e, PerkinElmer).⁴⁷ The Pd-SNP-PLA and CSNP-PLA-NFs (30 mg) were immersed in 30 mL of 1 $\times$ PBS buffer and incubated at room temperature. Aliquots consisting of 1 mL samples were taken at specific intervals and replaced with the fresh buffer to maintain a sink condition. Each aliquot was diluted 4-fold with a fresh buffer solution to assess the quantity of silver released at various times of up to 120 h. The release experiment was performed in triplicate, and average values were reported with respect to the percent release of silver ions. The release ratio of silver ions from Pd-SNP-PLA-NFs was measured on the basis of the released quantity of silver ions to the total amount of silver in Pd-SNP-PLA-NFs. The Ritger–Peppas equation was used to interpret the release kinetics of silver ions from Pd-SNPs-PLA-NF in 1 $\times$ PBS buffer.⁴⁸ The kinetic constant (k) was estimated from the following equation

$$\frac{M_t}{M_\infty} = kt^n \quad (4)$$

where k is the kinetic constant, M_t/M_∞ is the fractional release of the drug at time t , and n is the diffusion index (indicative of the Fickian and non-Fickian release behaviors) of silver ion release systems.

When $n < 0.5$ indicates a Fickian diffusion controlled-release, $0.5 < n < 1.0$ stands for non-Fickian (anomalous) transport and 1.0 means case II transport (zeroth order). $n > 1.0$ indicates supercase II transport.

Evaluation of the Antibacterial Activity of Pd-SNPs-PLA Nanofibers against *Listeria monocytogenes* during Cheese Storage. Commercial processed cheddar cheese (Amul, India) was purchased from the local market and was sliced into 1 cm $\times$ 1 cm $\times$ 1 cm pieces. The pieces were exposed to ultraviolet light for 15 min. Thereafter, these sliced pieces of cheese were spiked by dipping in 10^8 CFU/mL of *L. monocytogenes* in BHI broth. The spiked cheese slices were wrapped inside a Pd-SNP-PLA-NFs mat (on aluminum foil as the base material) and were incubated at both 4 and 37 $^\circ\text{C}$ for 6 days. The spiked cheese samples wrapped inside PLA nanofibers were taken as a control. The samples were taken out every 24 h and serially diluted in 1 $\times$ PBS and spread on a BHI agar plate to count the *L. monocytogenes* cells. The data plotted was CFU/g vs days.

RESULTS AND DISCUSSION

Synthesis and Characterization of Pd-SNPs. To demonstrate the potential of Pd-SNPs as a new class of antibacterial nanomaterials, we purified the biologically active pediocin peptide to homogeneity (Figure 1A) and used it to decorate Ag° using a single-pot chemical reduction method (Figure 1B) which involves the reduction of silver nitrate by sodium borohydride as a reducing agent, whereas pediocin acts as the stabilizing agent. Upon addition of sodium borohydride to the solution, a gradual change in the color of the reaction medium (containing pediocin and silver nitrate as precursor molecules) from colorless to brown revealed evidence of the synthesis of pediocin-capped silver nanoparticles (inset in

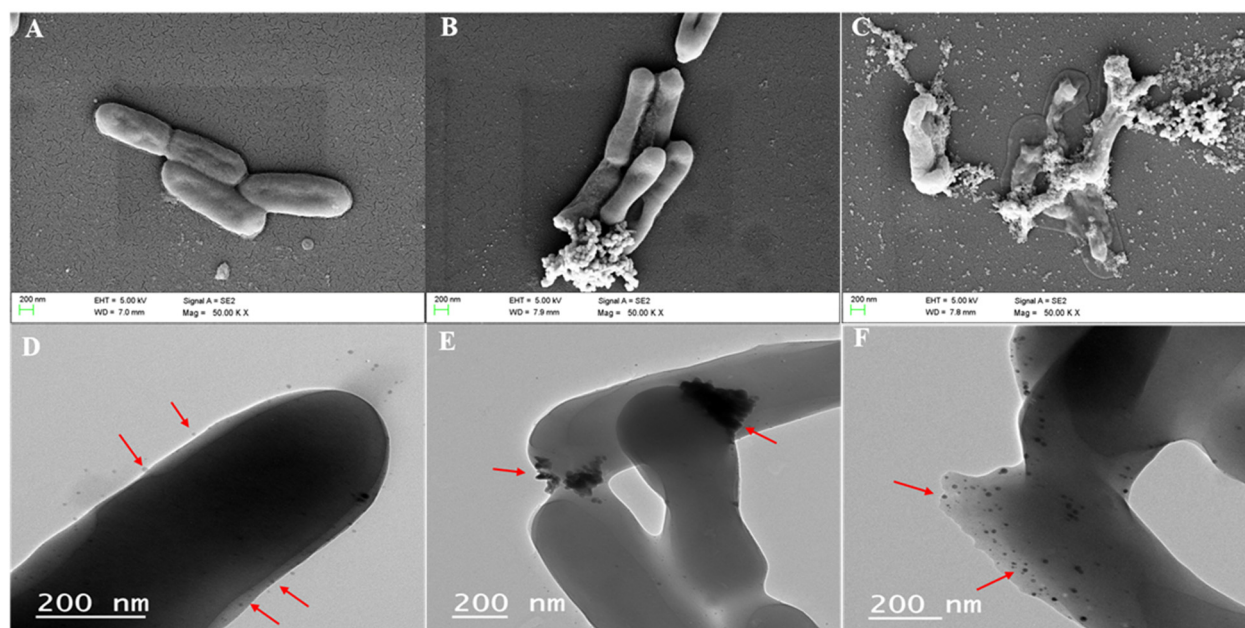


Figure 2. Pd-SNPs' induced morphological changes in *Listeria monocytogenes* cells were imaged by (A–C) FE-SEM and (D–F) HR-TEM. FE-SEM images of *L. monocytogenes* (A) without treatment, (B) CSNPs at the MIC value of Pd-SNPs, and (C) Pd-SNPs at MIC values. HR-TEM images of *L. monocytogenes* (D) at 0 min, (E) at 30 min, and (F) at 120 min of incubation with Pd-SNPs. The red arrows indicate the attachment of Pd-SNPs to *L. monocytogenes* cells.

Figure 1C). The borate anions generated during the reaction of borohydride with water adsorbed onto the surface and stabilized the silver colloid solution via the electrostatic barrier to aggregation.⁴⁹ The characteristic surface plasmon resonance peak of pediocin-capped silver nanoparticles was red-shifted from 390 nm (citrate-capped silver nanoparticles) to 405 nm, which qualitatively indicates the increase in the size of silver nanoparticles due to the capping of pediocin (Figure 1C).⁵⁰ Transmission electron microscopy (TEM) analysis demonstrated that Pd-SNPs were spherical in shape with a mean diameter of 22 nm, whereas CSNPs displayed a mean diameter of 15 nm (Figure 1D). The average hydrodynamic diameter (R_h) determined by dynamic light scattering (DLS) revealed that the biofunctionalized nanoparticles displayed an R_h of 107 nm (polydispersity index, PDI = 0.210, Figure S1) with a single peak in the size distribution curve which is in the acceptable limit of a narrow size distribution (PDI < 0.3) for various food and pharmaceutical formulations.^{51,52} The formation of larger nanoparticles substantiated the capping of pediocin onto the SNPs. The zeta potential (ζ) of the Pd-SNPs was observed to be slightly less (-25 ± 0.45 mV) as compared to that of CSNPs (-30 ± 0.6 mV) (Figure 1E). This surface potential neutralization and reduction of the negative charges of silver nanoparticles depends on the electrostatic interactions with the positive charges of the cationic peptide decorating the surface of SNPs.^{27,53,54} FTIR spectra of pediocin alone demonstrated the peak from 1600 to 1700 cm^{-1} that corresponds to amide-I (C=O stretching vibrations), the peak at 1271 cm^{-1} that relates to amide-III (N–H bending and C–N stretching vibrations; 1229–1301 cm^{-1}), the peak from 1000 to 1250 cm^{-1} that relates to the aliphatic amine (C–N stretching vibrations), and the peak from 3300 to 3500 cm^{-1} that relates to amide-A (axial O–H and N–H stretching vibrations) of the peptide (Figure 1F). The FTIR spectra of Pd-SNPs showed the broadening of IR peaks at 520 cm^{-1} (due to Ag–Ag/Ag–O vibrations)⁵⁵ and axial O–H and N–H stretching vibrations

at 3443 cm^{-1} (due to the involvement of the amino group in functionalization with SNPs) whereas, like pediocin, amide-I and amide-III peaks were found at the same positions in Pd-SNPs, confirming insignificant changes in the conformation of pediocin upon assembly at the surface of the nanobiocjugate and pediocin acting as a capping agent to prevent agglomeration. Furthermore, NMR analysis of Pd-SNPs revealed the broadening of the N–H resonances (Figure 1G), and aliphatic regions of the peptide resonances are intact in the biofunctionalized nanoparticles (Figure 1H), thus corroborating the involvement of amino groups in electrostatic interactions with SNPs. The biofunctionalized nanoparticles could not only facilitate the selective interaction with the negatively charged bacterial membrane but also assist direct intracellular translocation. In addition, the capping of pediocin on the nanoparticles provides a sustainable release of silver ions to the surrounding area. Furthermore, it is interesting that the synergistic interaction of SNPs with pediocin would also lead to a significant decrease in toxicity associated with silver ions, which is a critical criterion for biomedical applications.

Determination of the Minimum Inhibitory Concentration (MIC) and Fraction Inhibitory Concentration (FIC) Indexes. We examined the antibacterial efficacy of Pd-SNPs vis-à-vis CSNPs as the control by determining their minimum inhibitory concentrations (MICs) against two foodborne Gram-positive (*L. monocytogenes* and VISA-ST1745) and one Gram-negative multiple-antibiotic-resistant pathogen (*V. fluvialis* L-15318). Using pediocin to decorate SNPs resulted in superior antibacterial activity, with an ~ 13 times lower MIC (2.6 $\mu\text{g/mL}$) for *L. monocytogenes*, a 5 times lower MIC (2.5 $\mu\text{g/mL}$) for VISA-ST1745, and an ~ 10 times lower MIC (0.025 $\mu\text{g/mL}$) for *V. fluvialis* L-15318 as compared to the MIC of CSNPs. We calculated the fractional inhibitory concentration index (FICI) value to explore the synergistic interaction between pediocin and CSNPs. The FIC index of 0.389 for *L. monocytogenes* and 0.097 for *V. fluvialis* L-

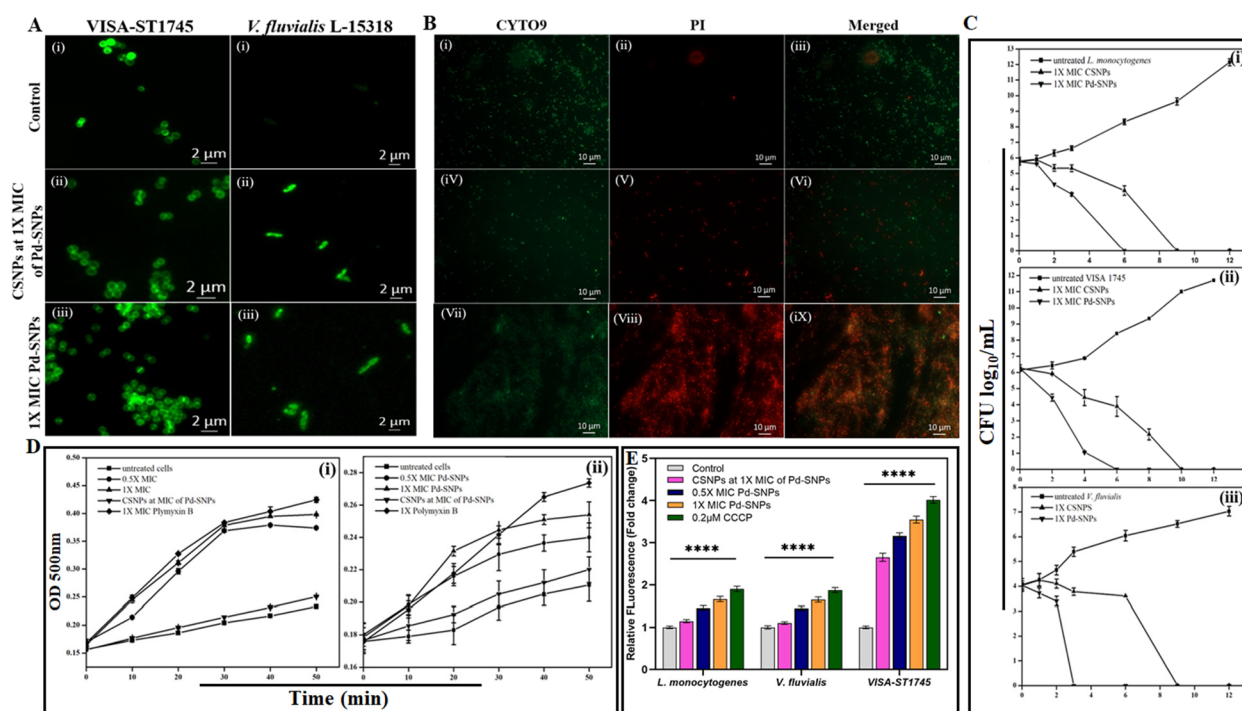


Figure 3. Mechanism of action studies of Pd-SNPs. (A) Confocal scanning laser microscopy (CSLM) images of the Pd-SNP-induced permeabilization of the membranes of both Gram-positive (VISA-ST1745) and Gram-negative (*V. fluvialis* L-15318) bacteria. (i) Untreated VISA-ST1745 (left) and *V. fluvialis* L-15318 cells (right) stained with BODIPY FL vancomycin (1 μ g/mL), (ii) CSNPs at the MIC of Pd-SNP-treated VISA-ST1745 (left) and *V. fluvialis* L-15318 cells (right) stained with BODIPY FL vancomycin (1 μ g/mL), and (iii) Pd-SNP-treated VISA-ST1745 (left) and *V. fluvialis* L-15318 cells (right) stained with BODIPY FL vancomycin (1 μ g/mL). (B) Live/dead staining-based microscopic visualization of *L. monocytogenes* cells. (i–iii) Untreated live, (iv–vi) CSNPs at the MIC of Pd-SNPs treated, and (vii–ix) 1X Pd-SNPs MIC treated *L. monocytogenes* cells. (C) Kill kinetic curves of (i) *L. monocytogenes*, (ii) VISA-ST1745, and (iii) *V. fluvialis* L-15318 cells. The number of bacterial colonies counted in the presence of (▲) CSNPs and (▼) Pd-SNPs. An untreated control sample (■) is also included. Error bars represent the standard deviation from the mean ($n = 3$). (D) Insertion of Pd-SNPs into the bacterial membrane, causing rapid outer and inner membrane permeabilization of *E. coli* ML-35p. Concentration- and time-dependent effects of Pd-SNPs on (i) the outer membrane were determined using the hydrolysis of nitrocefin by β -lactamase, and (ii) the effects on the inner membrane were determined using the hydrolysis of ONPG by β -galactosidase. Error bars represent the standard deviation from the mean ($n = 3$). (E) Concentration-dependent effect of Pd-SNPs on the membrane potential of *L. monocytogenes*, *V. fluvialis* L-15318, and VISA-ST1745 cells. Change in the membrane potential ($\Delta\Psi$) of bacterial cells due to the increased uptake of voltage-sensitive dye DiBAC4 and of DiSC3(5) upon treating Pd-SNPs. CCCP, a known ionophore, was taken as a positive control. Error bars represent the standard deviation from the mean ($n = 3$). One-way ANOVA, **** $P < 0.05$, shows the significant difference at 1X and 2X MIC and CSNPs at MIC of Pd-SNPs as compared to control cells. Error bars represent the standard deviation from the mean ($n = 3$).

15318 indicated that capping pediocin on SNPs exerted synergistic antibacterial activity against both pathogens (Table S1). Following this, we assessed the antibacterial spectrum of Pd-SNPs against a panel of multidrug-resistant microorganisms categorized as ESKAPE.² Pd-SNPs effectively inhibited ESKAPE pathogens at a concentration of 0.156 to 2.5 μ g/mL (Table S2). This result indicates that Pd-SNPs have the potential to combat a broad spectrum of pathogenic bacteria. Furthermore, scanning electron microscopy (SEM) (Figure 2A–C) and high-resolution transmission electron microscopy (HR-TEM) (Figure 2D–F) analysis revealed that the accumulation of Pd-SNPs on the bacterial cell surface caused extensive damage to the cell membrane integrity. Pediocin is known to target Gram-positive bacteria by interacting with the mannose phosphotransferase system (Man-PTS) as the membrane receptor,⁵⁶ whereas for Gram-negative bacteria, the enhanced antibacterial activity of Pd-SNPs could be ascribed to the increased local concentration of charges provided by the polyvalent effect of the cationic peptide on the biofunctionalized nanoparticles, which could lead to greater interaction with the bacterial membrane followed by

the leakage of cytoplasmic contents.^{57,58} To test this assumption, we followed the interaction of Pd-SNPs with membrane proteins of *V. fluvialis* L-15318 and *L. monocytogenes*. With the increasing concentration of Pd-SNPs, the fluorescence intensities of membrane proteins were significantly quenched (Figure S2). The presence of an upward-curving Stern–Volmer plot ($K_{sv} = 0.066$ for the Gram-negative membrane) indicated that the static quenching mechanism was involved in the binding of Pd-SNPs with membrane proteins.

Mechanism of Action Exploration with Biochemical, Biophysical, and Bioreporter Assays. To investigate whether the Pd-SNP interaction generates transient holes that lead to the opening of entry points through the outer membrane, we followed the localization of BODIPY-vancomycin in Pd-SNP-treated VISA-ST1745 and *V. fluvialis* L-15318, which displayed an increase in fluorescence at the surface and at the division septa of VISA cells (Figure 3A). In Gram-positive bacteria, the fluorescent vancomycin binds to its primary target lipid II, which is located at the division septum.⁵⁹ However, in the case of *V. fluvialis* L-15318, Pd-SNP treatment permeabilized the cells, as a result of which the

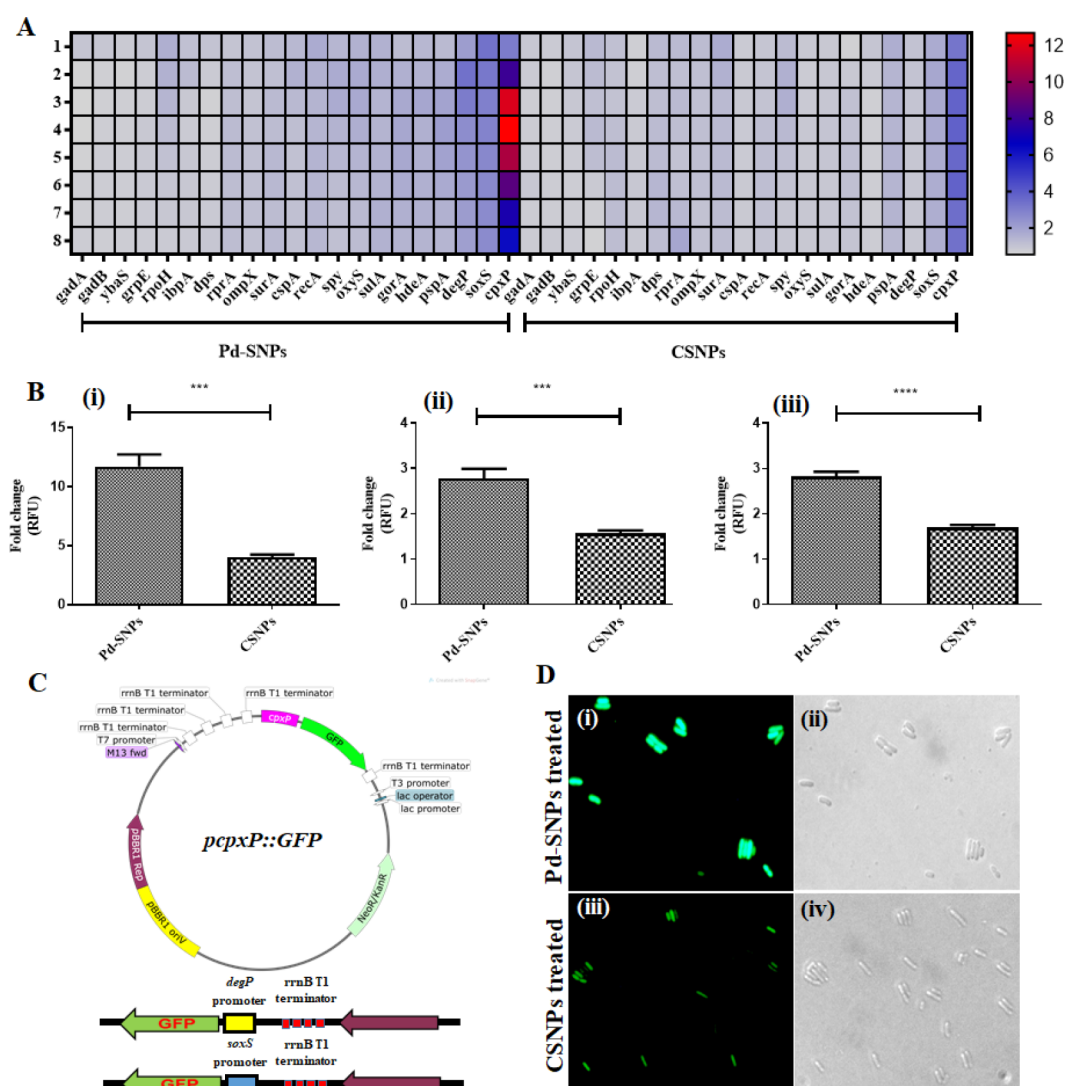


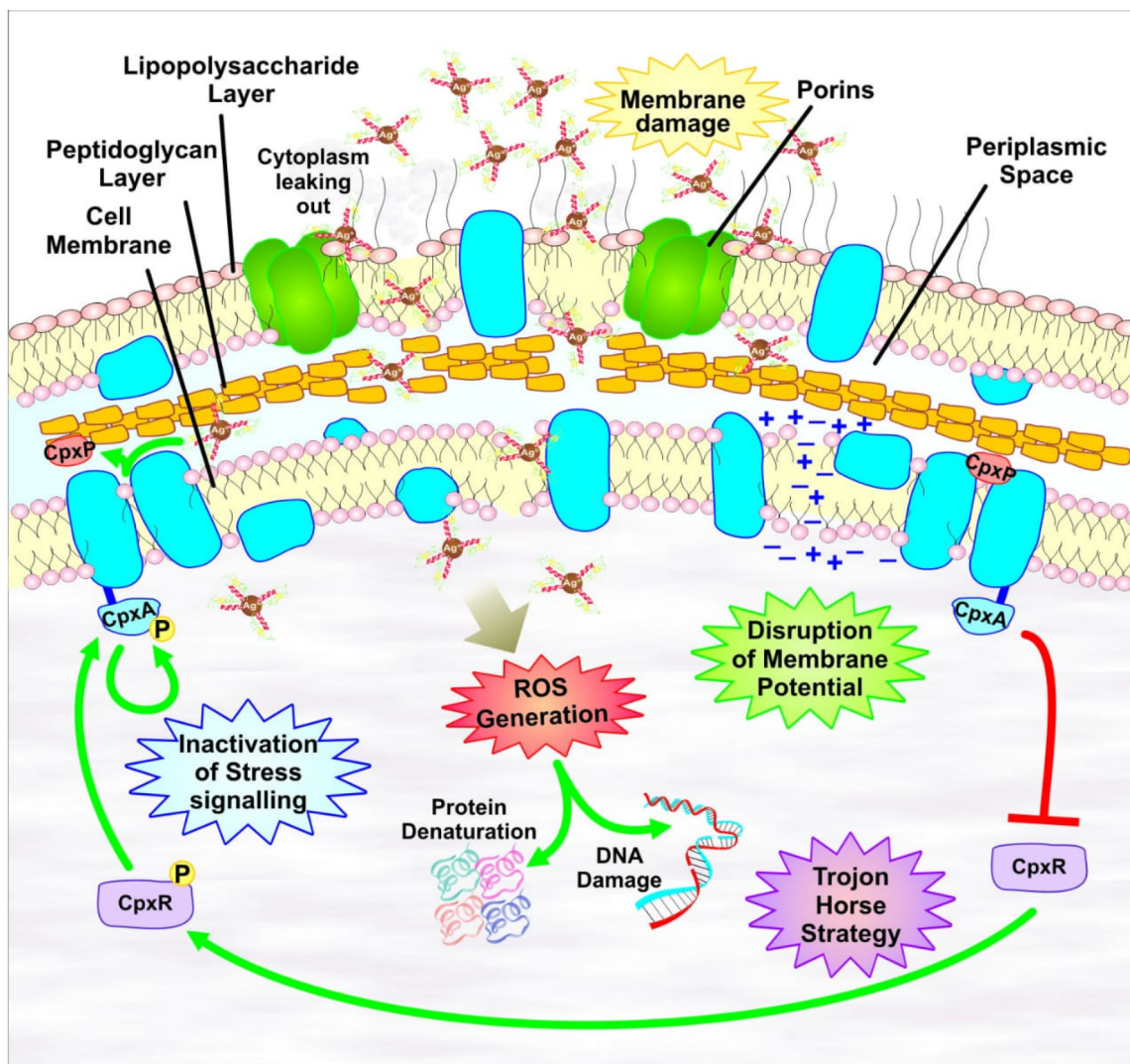
Figure 4. Bioreporter assay to deduce the mechanism of action of Pd-SNPs. (A) Heat map unveiling the temporal response of stress-responsive bioreporters in the presence of Pd-SNPs and CSNPs at 0.5× MIC. The GFP expression kinetics of stress-responsive bioreporters were depicted as fold-change values. (B) Response of selected *E. coli* bacterial bioreporters (i) *cpXP::GFP*, (ii) *degP::GFP*, and (iii) *soxS::GFP* to Pd-SNPs and CSNPs at a relative 0.5× MIC. Error bars represent the standard deviation from the mean ($n = 3$). The GFP expression of the selected bioreporters was represented as fold-change values. The results were analyzed using an unpaired *t* test (**** $P < 0.0001$; *** $P < 0.001$). (C) Construction of a plasmid map showing selected promoter sequences associated with stress-responsive signaling pathways cloned upstream to green fluorescent protein (GFP). (D) (i and ii) Fluorescent and DIC images of the *cpXP::GFP*-based bacterial bioreporter treated with 0.5× Pd-SNPs. (iii and iv) Fluorescent and DIC images of the *cpXP::GFP*-based bacterial bioreporter treated with 0.5× CSNPs.

bacterial cells were stained with BODIPY-vancomycin. This led us to conclude that Pd-SNPs cause the permeabilization of both Gram-positive and Gram-negative bacterial membranes and thus are able to exert a bactericidal effect.

A population of foodborne pathogens in the food matrix can enter a temporary viable but nonculturable bacterial (VBNC) state of existence that cannot be cultured in nutrient-rich laboratory media. Such VBNCs pose a substantial threat to public health and food safety because they cannot be detected by conventional microbiological methods.⁶⁰ An assessment of the effect of Pd-SNPs treatment on VBNCs by live/dead staining revealed that a majority of the Pd-SNP-treated *L. monocytogenes* cells were dead (PI-stained, red fluorescence). However, most of the cells were viable after treatment with CSNPs at the MIC of Pd-SNPs (SYTO 9-stained, green fluorescence), demonstrating the enhanced antibacterial effect of the nanobiotic material (Figure 3B).

Time-kill assays revealed that at 1× MIC there was a significant reduction of all tested bacterial pathogens (*L. monocytogenes*, VISA-ST1745, and *V. fluvialis*). Pd-SNPs inhibited multiantibiotic-resistant Gram-negative *V. fluvialis* L-15318 with >5 log reductions in CFUs within 3 h of the treatment (Figure 3C), and in the same time period, ~2 and 3.3 log reductions in CFUs were observed in the case of Gram-positive pathogens (*L. monocytogenes* and VISA-ST1745, respectively). These results indicate that at the MIC level, complete growth inhibition of both Gram-positive bacterial strains was achieved within 6 h, whereas the multiple antibiotic-resistant Gram-negative *V. fluvialis* was inhibited within 3 h. The efficient antibacterial activity of Pd-SNPs indicated that the capping of pediocin on SNPs was central to the superior performance of the biofunctionalized nanomaterial.

Scheme 1. Schematic Outlining the Mechanism of Action of Pd-SNPs Combating Pathogenic Bacteria



A real-time assessment of the outer and inner membranes upon Pd-SNP exposure to the *Escherichia coli* ML-35p strain revealed a dose- and time-dependent increase in the permeability of both membranes, accounting for the lethal effects (Figure 3D). These observations were confirmed further by examining their implications for the membrane potential of bacterial cells. The results show that Pd-SNPs caused ample depolarization and retained DiBaC4 and DiSC3(S) dyes, thus confirming that biofunctionalized nanoparticle treatment severely disrupted the membrane integrity (Figure 3E).

Antibacterial molecules are known to exert bactericidal effect through either ROS-dependent⁶¹ or ROS-independent pathways.⁶² We used the 2'-7'-dichlorofluorescein diacetate (DCFH-DA) dye to assess the role of reactive oxygen species (ROS) in the damage caused by Pd-SNPs. The generation of ROS triggered toxicity in all of the bacterial pathogens tested, confirming that the antimicrobial ability of the Pd-SNPs involves the generation of intracellular ROS (Figure S3A). These results implied that in the vicinity of the bacterial membranes, Pd-SNPs not only result in bacterial membrane perturbations but also elevate ROS production to trigger bacterial cell death. An assessment of DNA damage using a

terminal deoxynucleotidyl transferase-mediated dUTP nick end-labeling (TUNEL) study demonstrated that 79% of *L. monocytogenes* cells were TUNEL-positive after Pd-SNP exposure (Figure S3B), which could be reversed by thiourea (a ROS quencher). These results highlight the direct role of Pd-SNPs in DNA damage and the SOS response in *L. monocytogenes* through ROS production.

To explore the molecular mechanisms underlying the antibacterial effect of Pd-SNPs, we employed an array of genetically engineered *E. coli*-based bioreporters expressing the green fluorescent protein (GFP) (Figure 4A,C; Table S3). Bioreporters based on regulatory elements of global extracytoplasmic elements (*cpxP* and *degP*) and oxidative stress responses (*soxS*) showed a significant induction of GFP upon treatment with Pd-SNPs (11.6-, 2.7-, and 2.8-fold, respectively) in comparison to a control, whereas CSNPs at sub-MIC concentration showed 3.9-, 1.5-, and 1.7-fold increases in the fluorescence intensity for *cpxP*, *degP*, and *soxS*, respectively (Figure 4B,D). *cpxP*, a periplasmic chaperone, interacts with *cpxA* and keeps stress signaling in a resting state.⁶³ Pd-SNPs activate *cpxP* and thus keep the bacterial cell oblivious to the stress exerted. As a result, newly synthesized proteins cause the enhanced killing of the Pd-SNP exposed cells.⁶⁴ Activation of

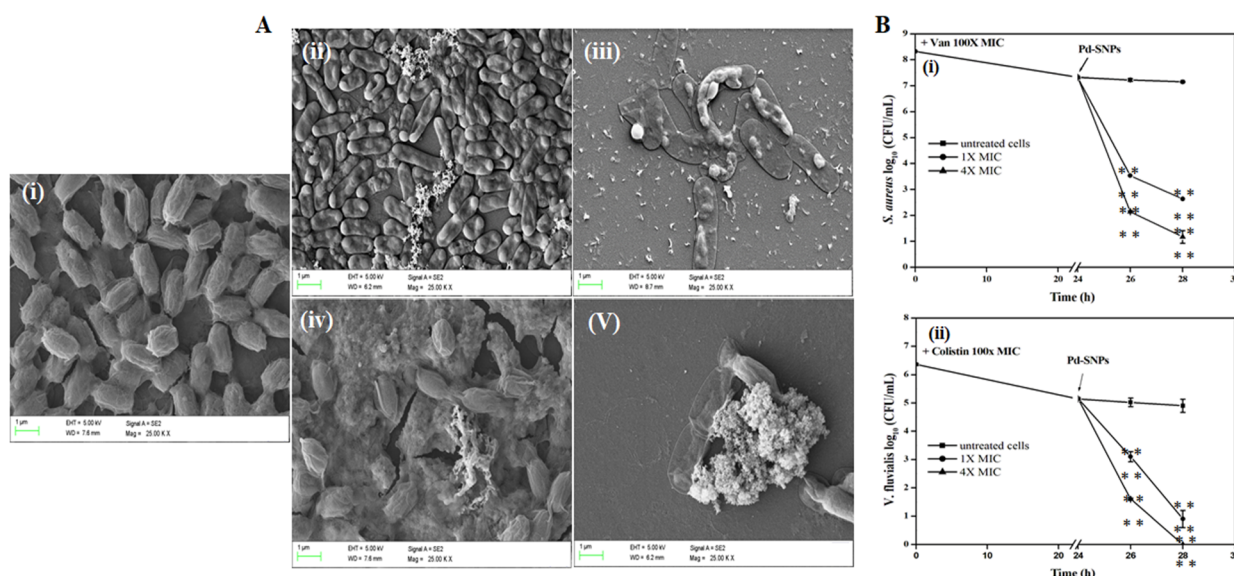


Figure 5. Antibiofilm and antipersister activity of Pd-SNPs. (A) Scanning electron microscope images of (ii and iii) biofilm inhibition and (iv and v) eradication by Pd-SNPs. (i) Untreated *L. monocytogenes* showing a biofilm of the bacteria, (ii) CSNPs at the MBIC of Pd-SNPs, (iii) MBIC of Pd-SNPs added prior to biofilm formation, (iv) CSNPs at 5× MBIC of Pd-SNPs, and (v) 5× MBIC of Pd-SNPs added after biofilm formation. (B) Bactericidal activity of Pd-SNPs against persister cells. (i) *S. aureus* (VISA-ST1745) cells were treated for 24 h with 100× MIC of vancomycin, and (ii) *V. fluvialis* L-15318 cells were treated for 24 h with 100× MIC of colistin. Antibiotics were removed (arrow), and persister cells were exposed to PBS containing 1× and 4× MIC of Pd-SNPs or no NPs for an additional 2 or 4 h. Error bars represent the standard deviation from the mean ($n = 3$). One-way ANOVA, **** $P < 0.0001$, shows a significant difference from the untreated control after 26 and 28 h of treatment, respectively.

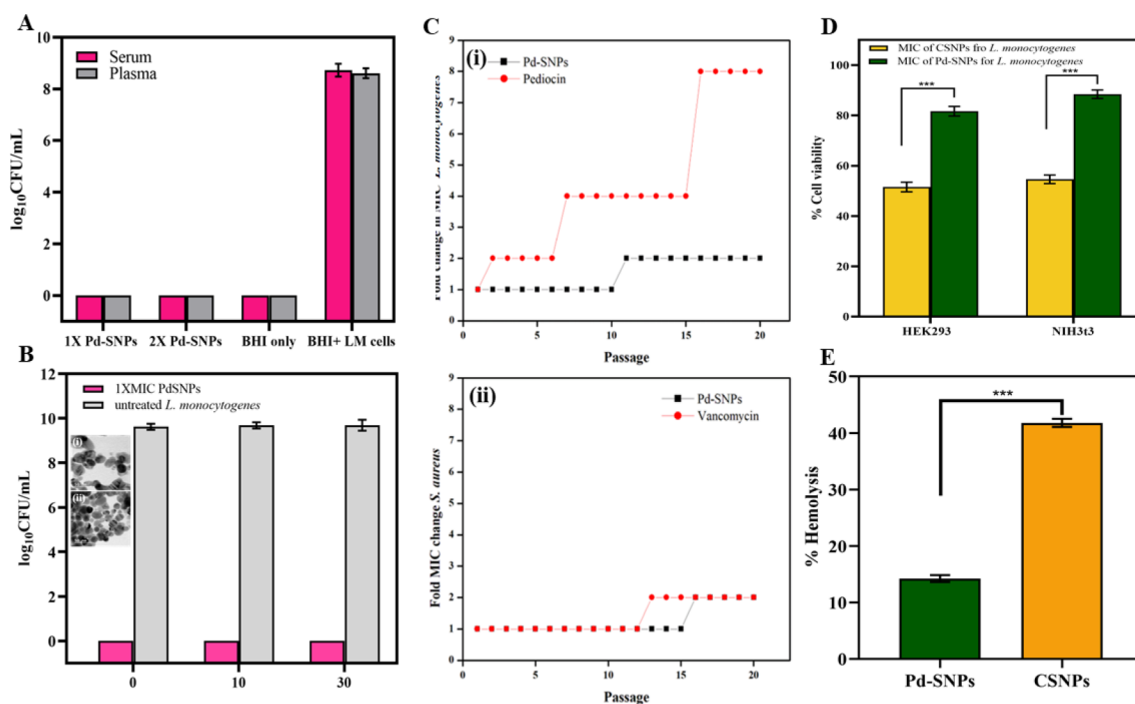


Figure 6. Biocompatibility, thermostability, resistance evolution, and toxicity assessment of Pd-SNPs. (A) Antilisterial activity of Pd-SNPs in human serum and plasma at 1× and 2× MIC levels. BHI represents the brain–heart infusion medium, and LM represents *L. monocytogenes*. For a representation of the data on a logarithmic scale, a value of 1 CFU was assigned to no growth of bacteria. (B) Effect of temperature on the morphology and antilisterial activity of Pd-SNPs. Antilisterial activity of Pd-SNPs before and after incubation for 10 and 30 min at 100 °C. The inset shows TEM images of (i) untreated and (ii) heat-treated Pd-SNPs. The data represents an average of three replicates $\pm$ the standard deviation shown by the error bars. (C) Resistance acquisition of *L. monocytogenes* and VISA-ST1745 against Pd-SNPs. (i) Resistance development of *L. monocytogenes* to Pd-SNPs and pediocin. (ii) Resistance development of VISA-ST1745 to Pd-SNPs and vancomycin. Values are magnitude changes in the MIC relative to the MIC of the first passage. (D) Cell viability of HEK293 and NIH3T3 at their respective 50× MICs of Pd-SNPs and CSNPs against *V. fluvialis* L-15318. (E) Hemolysis (%) at respective 50× MICs of Pd-SNPs and CSNPs against *V. fluvialis* L-15318. Error bars represent the standard deviation from the mean ($n = 3$).

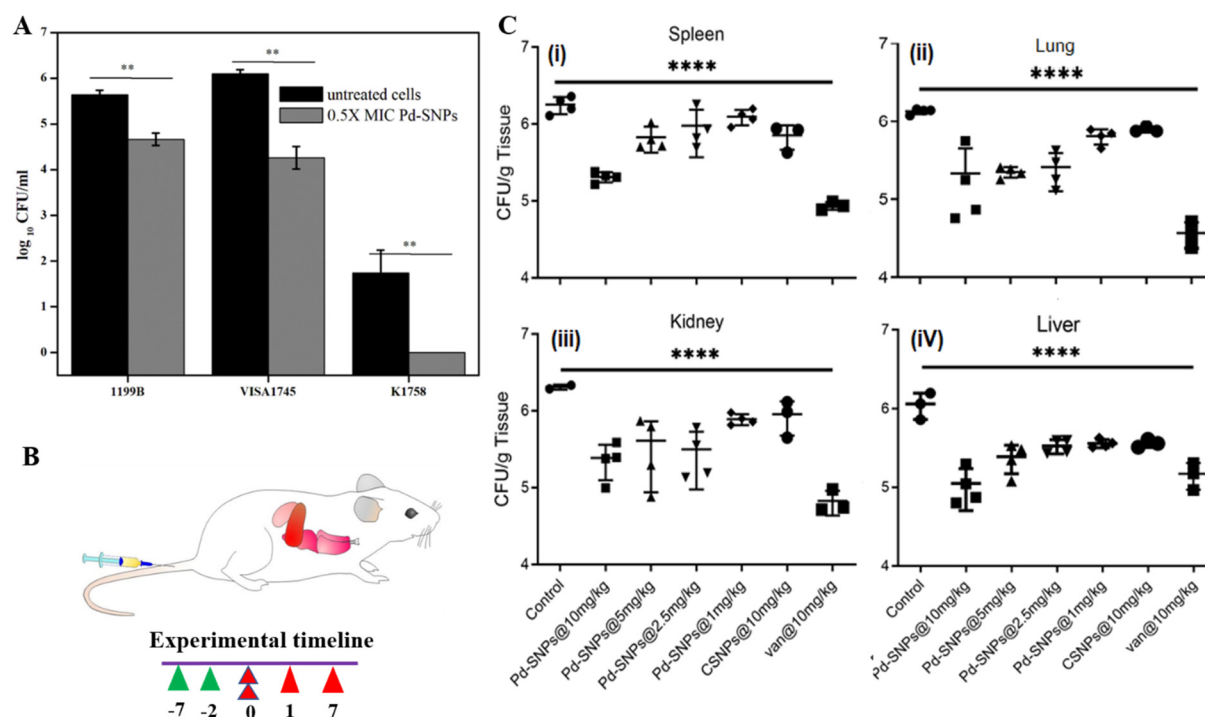


Figure 7. Assessment of *in vitro* invasiveness and *in vivo* efficacy of Pd-SNPs. (A) Invasiveness of *S. aureus* VISA-ST1745 in the presence of Pd-SNPs in HEK293 cells. Error bars represent the standard deviation from the mean ($n = 4$). A paired t test was performed to calculate the significant difference ($P < 0.05$). To represent the data on a logarithmic scale, a value of 1 CFU was assigned to no growth of bacteria. (B) Schematic of the experimental design showing time scales of infection with VISA-ST1745. The female Swiss albino mice were acclimatized for 7 days (denoted as -7) and randomly grouped 2 days before (denoted as -2) the start of the experiment. The mice were divided ($n = 6$) into a negative control (vehicle), a positive control (vancomycin 10 mg/kg intravenous), and treatment groups (doses of 1, 2.5, 5, and 10 mg/kg of body weight of Pd-SNPs) as well as CSNPs at a higher dose of 10 mg/kg. At day 0 (at the start of experiment), mice were infected with a bacterial strain via an intravenous route with 1×10^6 CFU/mL of VISA-ST1745 and 24 h postinfection (denoted as 1). Pd-SNPs treatment (oral administration) was given for 7 days postinfection (denoted as 7), following which the animals were euthanized and vital organs including the spleen, kidneys, liver, and lungs were collected. (C) *In vivo* efficacy of Pd-SNPs in a mouse (VISA-ST1745 infected) model. CFU of VISA-ST1745 found in the (i) spleen, (ii) lungs, (iii) kidneys, and (iv) liver of infected mice 24 h after treatment with various concentrations of Pd-SNPs (1, 2.5, 5, and 10 mg/kg) or vancomycin (10 mg/kg). All data are expressed as the mean $\pm$ standard deviation as indicated by the error bars. **** $P < 0.0001$, a one-way ANOVA test was performed to calculate the significant difference as compared to the untreated control.

the *soxS* bioreporter indicates that oxidative stress is generated upon exposure of Pd-SNPs to bacterial cells.⁶⁵

Taken together, the genetic and biochemical data demonstrate that through the induction of the *cpxP* pathway, Pd-SNPs utilize a Trojan horse approach to inflict lethal damage on bacterial cells, which is further compounded by *degP* gene upregulation. This genetic interplay leads to extracytoplasmic stress in the bacterial pathogens, leading to ROS generation with lethal consequences. Overall, the mechanism of action of Pd-SNPs is illustrated in Scheme 1.

Antibiofilm and Antipersister Activity of Pd-SNPs.

Since Pd-SNPs can express bactericidal activity even at very low concentrations, we investigated the mitigation ability of the Pd-SNPs against biofilm formation and eradication using an *in vitro* biofilm model. Biofilm formation in bacterial pathogens is one of the mechanisms for avoiding antibacterial therapies, enhancing adherence to the host cells, and thus becoming more virulent.⁶⁶ Microbes growing within biofilms become extremely resistant to antimicrobials as compared to their planktonic counterparts.⁶ On the basis of CV staining, the minimum biofilm inhibitory concentration (MBIC) of Pd-SNPs against *L. monocytogenes* was calculated to be 20.8 $\mu\text{g/mL}$ (8 $\times$ MIC against planktonic cells, Figure S4B), which was 6.4-fold lower than the MBIC of CSNPs (133.2 $\mu\text{g/mL}$, Figure S4A). For *V. fluvialis* L-15318, the MBIC of Pd-SNPs was

20.8-fold lower (0.05 $\mu\text{g/mL}$; 2 $\times$ MIC of planktonic cells, Figure S4D) as compared to that of CSNPs (1.04 $\mu\text{g/mL}$, Figure S4C). SEM was performed to examine the morphological effects of the Pd-SNPs on the biofilm of *L. monocytogenes*. The SEM micrograph of the biofilm revealed that Pd-SNPs prevented the formation of biofilms and led to their dispersion (Figure S4iii). In contrast, a biofilm treated with CSNPs at the MBIC of Pd-SNPs showed intact dense clusters of bacteria that remained intact and lacked any apparent loss of membrane integrity. Furthermore, we challenged the established biofilm of *L. monocytogenes* with Pd-SNPs and discovered that 5 $\times$ MIC of Pd-SNPs was required to eradicate a similar amount of biofilm (Figure S4v). The results that are presented highlight the strong antibiofilm properties of the Pd-SNPs.

To assess the effectiveness of Pd-SNPs against metabolically dormant persisters, we isolated VISA-ST1745 and *V. fluvialis* L-15318 persister cells against clinical frontline antibiotics vancomycin and colistin, respectively. Once the antibiotic pressure is removed, these persister cells are known to be a source of recurrent infections.⁶⁷ Treatment of *V. fluvialis* L-15318 persister cells with Pd-SNPs at the 4 $\times$ MIC level led to the complete inhibition of target persister cells, whereas for VISA-ST1745, 7 log reductions in CFUs were observed, showing that Pd-SNPs can successfully mitigate persister cells

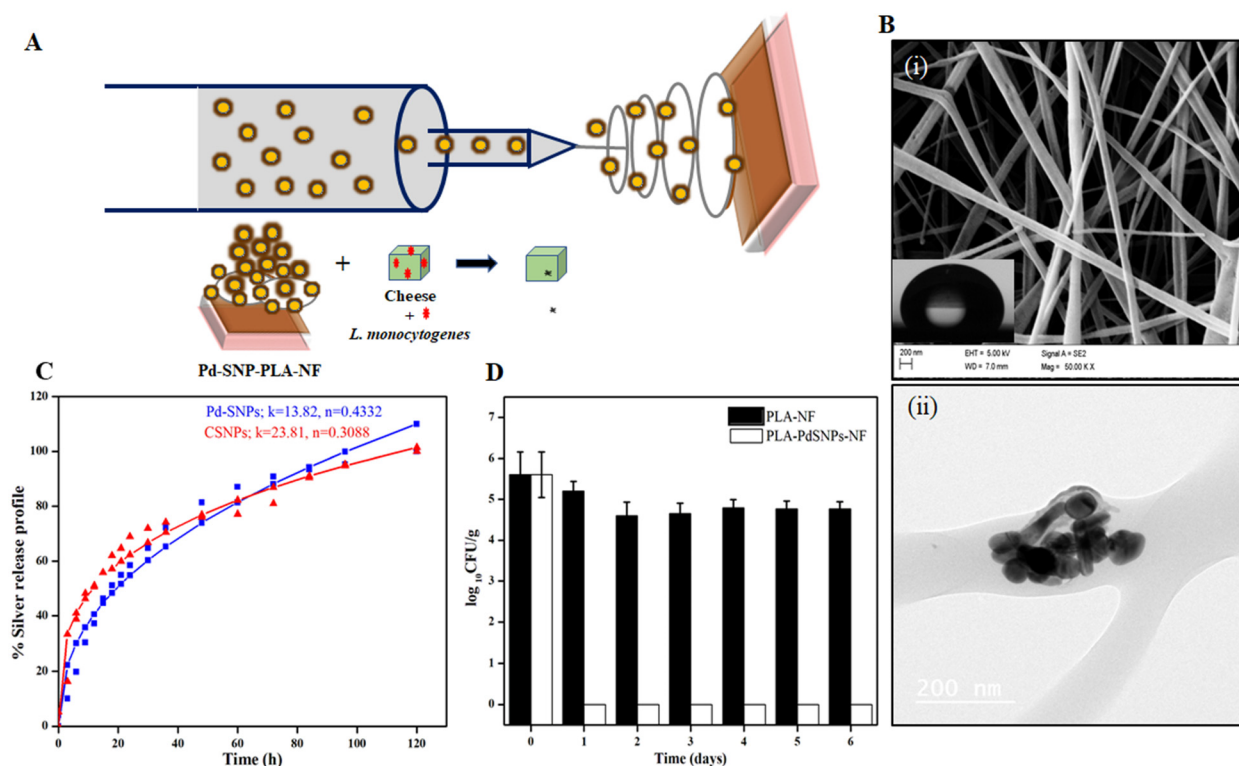


Figure 8. (A) Schematic of the synthesis of Pd-SNPs-PLA nanofibers. (B) Images of Pd-SNPs-PLA-NF via (i) FE-SEM and (ii) TEM. The inset shows the contact angle of Pd-SNPs-PLA-NFs. (C) Release profile of silver ions fitted with the Ritger–Peppas equation. (D) Total bacterial counts in wrapped cheese samples with fabricated Pd-SNPs-PLA nanofibers during storage at 37 °C. For a representation of the data on a logarithmic scale, a value of 1 CFU was assigned to no growth of bacteria.

and exhibit promise to be used as a potential tool to mitigate the spread of antibiotic-resistant bacteria (Figure 5B). The potency of Pd-SNPs against persister cells can be attributed to the ability of Pd-SNPs to target membranes of the bacterial pathogens. Indeed, targeting the bacterial membrane function has been promoted as one of the methods to mitigate persistent bacterial infections.⁶⁷

Serum, Plasma, and Heat Stability Analysis of Pd-SNPs. Besides antipersister activity, Pd-SNPs efficiently inhibited the *L. monocytogenes* cells in serum and plasma at the same MIC values as those in a bacteriological medium, indicating that the nanobiotic system was stable and effective under physiological conditions (Figure 6A). Pd-SNPs also retained their antibacterial activity after being preheated to 100 °C for 30 min (Figure 6B). In general, AMPs are susceptible to proteases and show poor pharmacokinetics, leading to their limited exploration as antibacterial agents in clinical settings.¹⁸ The serum stability and amenability to sterilization are two desirable properties which make Pd-SNPs an attractive option for healthcare and food-packaging applications.

Evaluation of Resistance Development in the Presence of Pd-SNPs. One of the attributes of a potent arsenal against MDR pathogens is long-lasting antibacterial efficacy. We investigated the ability of *L. monocytogenes* and VISA-ST1745 to acquire resistance against Pd-SNPs. Following 20 repeated passages (over 250 generations) of *L. monocytogenes* cells to the MIC levels of Pd-SNPs, no significant change in the MIC of Pd-SNPs was reported (2.6 μg/mL at the first passage and 5.2 μg/mL at the last passage), while for VISA-ST1745 only a 2-fold increase in the MIC of Pd-SNPs after 15 passages (300 generations) was observed,

indicating that pathogens failed to evolve resistance against Pd-SNPs (Figure 6C). On the other hand, resistance to pediocin evolved quickly (8× MIC after 20 passages). The capping of pediocin on SNPs significantly delays the development of antibacterial resistance, which underscores the potential of Pd-SNPs for further clinical applications.

Assessment of Biocompatibility and Pathogen Invasiveness. The biocompatibility of Pd-SNPs was investigated by exposing the HEK293 and NIH3T3 cells and the hRBCs to varying concentrations of SNPs (Figure S5). Hemotoxicity and cytotoxicity are considered to be hurdles to the application of cationic AMPs in clinical settings.¹⁷ At MIC values, Pd-SNPs were nonhemolytic and noncytotoxic, whereas at 50× MIC (*V. fluvialis* L-15318), >90% of HEK293 cells and >92% of NIH3T3 cells were viable. At the same magnitude of MIC values, CSNPs exerted a high level of toxicity toward these cell lines (Figure 6D). Pd-SNPs exhibited a similar safety profile against hRBCs (<15% hRBC lysis at 50× MIC), whereas CSNPs showed toxicity (>40% hRBC lysis) (Figure 6E). Pd-SNPs exhibited a therapeutic index (cytotoxicity IC₅₀/MIC) of >20 for both Gram-positive pathogens, whereas for Gram-negative pathogen *V. fluvialis* L-15318, a therapeutic index of >2000 was calculated (Table S4). The values of the therapeutic index against all tested bacterial strains indicate that Pd-SNPs possessed selectivity toward bacterial cells compared to cell lines and hRBCs. This result establishes the biocompatibility of Pd-SNPs with mammalian cells and hRBCs.

Furthermore, the invasiveness of VISA-ST1745 cells was evaluated in the presence of 0.5× MIC using HEK293 cell lines. The results revealed that Pd-SNPs reduced the cellular

invasion of VISA-ST1745 and NorA-overproducing *S. aureus* SA-1199B by 2 log as compared to the control knockout strain (SA-K1758) (Figure 7A), which shows that Pd-SNPs are able to reduce the virulence and invasiveness of both the clinical and fluoroquinolone-resistant *S. aureus*.

In Vivo Assessment of Pd-SNPs in the VISA-ST1745 Infection Mouse Model. Enhanced antibacterial ability and favorable biocompatibility of Pd-SNPs led us to examine the *in vivo* antibacterial efficacy using a model of a Swiss albino mouse infected with VISA-ST1745 cells (Figure 7B). Up to a 1.2–1.5 log reduction in CFU was observed in different tissues ($p < 0.0001$) after the oral administration of Pd-SNPs (10 mg/kg of body wt), whereas mice treated with CSNPs (10 mg/kg of body wt) showed a 1 log reduction in CFU as compared to the vehicle control (Figure 7C). Clinically used frontline antibiotic vancomycin (10 mg/kg of body wt) showed up to a 2 log reduction in CFU ($p < 0.0001$) as compared to the vehicle control. The toxicity assessment of the Pd-SNPs at various oral dosages (25 mg/kg to 50 mg/kg of body wt) did not show any sign of illness or mortality throughout the experimental period. There was no obvious change in hematological parameters or serum biochemical parameters except for an increase in the cholesterol level, further indicating the hemocompatibility of Pd-SNPs (Table S5). Considering its impressive biosafety and *in vivo* bactericidal activity, Pd-SNPs show promise to be further developed into clinical antibacterial agents to combat general antibiotic-resistant infections and specific VISA infections. However, further studies to follow up on these observations need to be conducted.

Antibacterial Effectiveness of Biodegradable Pd-SNP-PLA Nanofibers on Cheese Samples Spiked with *Listeria monocytogenes*. We assessed the ability of biofunctionalized nanoparticles to mitigate foodborne pathogens as bioactive and biodegradable food packaging materials by incorporating Pd-SNPs into nanofibers of polylactic acid (PLA-NFs, FDA-approved biodegradable food-contact polymer⁶⁸ (Figure 8A) and characterized them (Figures 8B and S6A) and then explored the possibility of Pd-SNP-PLA-NFs in controlling *L. monocytogenes*.⁶⁹ Pd-SNP-PLA-NFs retained their antibacterial activity and inhibited *L. monocytogenes in vitro* (Figure S6A, inset). Silver ions were released from Pd-SNP-PLA-NFs with an initial burst within the first 18 h, followed by a continuous slow and sustained release for up to 120 h. The diffusion index (n) calculation revealed that the Fickian diffusion model was applicable to the release kinetics of silver ions from nanofibers.⁴⁸ The kinetic constant (k) for silver ion release was 1.72-fold higher for CSNP-PLA-NFs ($k = 23.81$) than for Pd-SNP-PLA-NFs ($k = 13.82$), showing the slow and sustained release of silver ions from Pd-SNP-NFs (Figure 8C). The sustained release of silver ions from the nanofibers was further corroborated by the fact that the contact angle for Pd-SNP-PLA-NF was $122.8 \pm 0.35^\circ$ as compared to $102.16 \pm 0.73^\circ$ for CSNP-PLA-NF.⁷⁰ To assess the suitability of such antibacterial material in food packaging, *L. monocytogenes*-spiked commercial cheese samples were wrapped inside the Pd-SNP-PLA-NFs coated on aluminum foil. Pd-SNP-PLA-NFs reduced the listerial counts by ~ 4 log after 6 days of storage at 37 °C (Figure S6B). However, a dramatic reduction in counts was observed (no viable listerial cells) after 6 days of storage at 4 °C (Figure 8D). Pd-SNP-PLA-NFs are biodegradable and can find potential applications in active food packaging to control foodborne pathogens using environmentally benign technologies.

CONCLUSIONS

We have engineered a highly effective and biocompatible nanobiotic system with versatile capability by decorating a food-grade antibacterial peptide with a narrow spectrum of activity on the silver nanoparticles. Decorating silver nanoparticles with pediocin peptide led to a remarkable enhancement in the spectrum and the antibacterial activity (including ESKAPE pathogens), with robust synergistic interaction leading to a more than 10-fold higher activity than for either of the individual components. The nanobiotic platform proved to be extremely heat-stable, nonhemolytic, noncytotoxic and could exert antibacterial potency in serum and plasma against multiantibiotic-resistant pathogens. The Pd-SNPs inhibited biofilm formation, effectively eradicated preformed biofilms, and mitigated persister bacterial pathogens effectively. The ability of Pd-SNPs to induce transient holes to open the entry point in nontarget Gram-negative cells along with the sequestration of *cpxP*-mediated stress signaling renders the bacterial pathogens vulnerable to ROS-induced collateral damage, eventually leading to cell death. *In vivo* studies in vancomycin intermediate *S. aureus*-infected mice showed that the nanobiotic system holds significant promise as a therapeutic modality in combating vancomycin-resistant bacterial infections. The GRAS status of the pediocin is an additional advantage in extending its applications in the food sector. We formulated the antibacterial packaging material by trapping Pd-SNPs in biodegradable nanofibers, which significantly reduced the listerial load in perishable food. We provide a stable vehicle with versatile antibacterial properties which goes beyond alleviating the extant concerns about the toxicity of developed nanoformulations. Taken together, these features suggest that Pd-SNPs hold promise in being used as a nonantibiotic nanoplatform for mitigating infectious bacterial pathogens and open up new avenues for their mitigation with “functional” properties in myriad applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.2c01385>.

Stern–Volmer plots, ROS generation studies, TUNEL assay showing DNA damage, biofilm inhibition study, dose-dependent hemolysis and cytotoxicity analysis, FT-IR analysis of Pd-SNP-PLA-NF, antilisterial activity of Pd-SNPs-PLA-NF in cheese samples, MIC spectrum against ESKAPE pathogens, FIC-index analysis, genetically engineered strains used for bioreporter analysis, therapeutic index, and toxicity analysis in the mouse model (PDF)

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Author Contributions

N.K.N. and P.K. conceptualized the research. P.K., A.A.S., P.K., V.K.G., R.D., T.K.S., K.G., and A.H. performed the research. P.K., K.N.R., R.K.M., K.M.P., R.P., and N.K.N. analyzed the data. N.K.N. and P.K. wrote the paper. All of the authors contributed to the discussions during the entire study.

Notes

The authors declare no competing financial interest.

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